

WITBANK, South Africa

WMO: 682730

Lat: 25.8338S

Lon: 29.1926E

Elev: 5102

StdP: 12.18

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.8	39.1	8.7	10.4	59.0	12.7	12.7	57.2	18.0	57.8	16.4	56.5	6.1	190	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	87.5	61.4	85.0	61.1	82.9	61.0	67.6	79.3	66.6	77.2	65.6	75.6	8.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	110.0	70.8	63.3	105.5	70.0	62.5	102.6	69.4	35.3	79.5	34.4	77.4	33.5	75.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.5	13.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			28.0	68.9	2.1	1.5	26.5	70.0	25.3	70.9	24.1	71.7	22.6	72.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	69.4	69.5	67.3	63.0	59.0	54.5	53.8	58.3	64.7	66.3	67.1	69.1		
	DBStd	7.23	3.82	3.55	4.17	4.08	4.75	4.30	4.11	5.75	6.18	5.52	5.06	4.42		
	HDD50	39	0	0	0	0	5	12	13	8	1	0	0	0		
	HDD65	1368	7	6	24	82	190	316	346	215	79	52	36	15		
	CDD50	4950	601	545	536	390	284	147	132	264	442	504	512	593		
	CDD65	803	143	131	95	22	4	0	0	6	71	91	98	142		
	CDH74	5548	851	703	496	136	74	6	3	107	669	856	765	882		
	CDH80	1222	208	144	75	13	5	0	0	2	154	233	176	212		
(14) Wind	WSAvg	6.3	6.7	6.6	6.2	5.5	5.1	5.6	5.6	6.0	7.0	7.5	7.2	6.5		
(15) Precipitation	PrecAvg	28.1	5.0	3.6	3.5	1.4	0.6	0.2	0.1	0.2	0.7	3.1	4.4	5.2		
	PrecMax	42.7	9.8	11.8	7.9	4.7	2.9	1.2	0.3	1.2	3.7	6.5	8.2	9.6		
	PrecMin	17.8	2.2	0.4	0.9	0.0	0.0	0.0	0.0	0.0	0.0	1.2	1.2	2.4		
	PrecStd	6.4	2.1	2.4	2.2	1.2	0.7	0.3	0.1	0.4	0.9	1.5	2.0	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.2	87.8	85.6	82.1	79.2	74.4	73.6	80.1	87.9	89.4	89.5	88.5		
		MCWB	64.5	63.1	60.3	58.8	54.3	51.0	48.1	53.9	56.8	60.1	59.8	65.1		
	2%	DB	85.4	84.2	82.2	77.9	75.6	70.7	70.4	77.3	84.3	85.6	84.9	85.1		
		MCWB	64.3	63.3	60.5	58.0	53.2	49.4	48.2	52.5	55.9	58.2	60.3	64.0		
	5%	DB	82.1	81.3	79.3	75.1	72.9	67.9	68.1	74.6	81.0	82.4	81.8	82.5		
		MCWB	64.7	63.6	61.1	57.8	52.8	48.3	47.4	51.2	54.4	57.7	60.5	63.9		
	10%	DB	78.8	78.6	76.8	72.3	69.8	64.9	65.2	71.3	77.9	78.9	78.7	79.2		
		MCWB	64.1	63.2	61.3	57.1	52.3	47.9	47.3	50.1	53.4	57.5	60.7	63.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	69.2	67.0	64.0	57.6	54.0	52.2	57.0	60.5	64.2	66.5	68.2		
		MCDB	81.7	81.2	75.9	74.1	71.7	68.8	65.5	76.1	78.6	80.8	79.0	82.0		
	2%	WB	67.7	67.3	65.4	62.5	56.0	52.0	50.9	55.0	58.6	62.2	64.6	66.9		
		MCDB	79.1	78.3	74.2	70.9	69.8	65.3	64.5	72.2	76.3	77.4	76.7	79.7		
	5%	WB	66.5	66.0	64.3	61.0	54.8	50.6	49.6	53.5	57.2	61.0	63.4	65.7		
		MCDB	76.1	76.0	72.5	68.5	68.4	63.1	63.4	69.2	75.0	74.8	75.5	77.4		
	10%	WB	65.5	64.8	63.1	59.4	53.5	49.4	48.3	51.9	55.8	59.7	62.3	64.7		
		MCDB	74.4	74.0	71.5	67.9	66.6	61.3	62.6	67.2	71.8	72.3	74.3	75.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.2	18.0	19.1	22.1	22.8	23.5	24.6	24.4	22.9	20.3	18.2		
		MCDBR	23.8	23.1	22.9	22.9	25.4	25.7	26.3	26.4	26.8	27.2	24.9	22.9		
		MCWBR	8.3	7.2	7.8	8.6	10.3	11.4	11.5	10.6	9.7	9.2	8.9	7.8		
	5% WB	MCDBR	18.4	19.2	17.4	18.8	22.8	22.4	23.6	24.6	25.2	22.8	21.0	19.5		
		MCWBR	7.7	7.4	7.2	8.0	9.7	10.7	11.1	10.3	9.8	9.2	8.8	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.390	0.390	0.380	0.335	0.298	0.279	0.281	0.336	0.380	0.416	0.378	0.385		
	taud		2.322	2.325	2.332	2.434	2.507	2.529	2.511	2.346	2.246	2.184	2.327	2.326		
	Ebn at Noon		303	298	291	290	289	289	292	285	286	286	304	305		
	Edh at Noon		44	43	40	33	28	26	27	36	43	49	43	44		
(44) All-Sky Solar Radiation	RadAvg		2094	2003	1765	1513	1389	1261	1367	1595	1878	2005	2091	2125		
	RadStd		202	183	157	130	69	63	31	73	92	126	182	121		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page